Continuous Curvature Turns Based Method for Least Maximum Curvature Path Generation of Autonomous Vehicle

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***Abstract*—Safe driving and stable paths are essential for the autonomous vehicle navigation, that large and fast steering angle should be avoided. This paper addresses a local path generation problem for autonomous vehicles while the least maximum curvature and the shortest length are obtained with limited curvature rate. The properties of the novel feasible paths based on continuous curvature turns are investigated. And a simple and fast computational method is presented to solve the problem by iterative procedure. The simulation in this paper shows that, compared with recent researches, the performance of least maximum curvature and shortest length is obtained by the novel local path planner, and the driving behavior is closer to the human driver operation.**

Keywords—trajectory generation, local path planning, autonomous vehicle, continuous curvature.

I. Introduction

Autonomous vehicle has been studied for several decades and the performance at the level of human race driver has been achieved on the structured road in recent years [1]. Path planning is one of the most important research topics in the autonomous vehicle and it can be categorized as global and local. The global path planning algorithm generates a series of collision-free waypoints and connects the original state with the target state. The local path planning algorithm focuses on the short-distance range and generates a trajectory between two or several waypoints. Furthermore, the kinematic constraints and the constrains from the environment are satisfied by the local path planner.

The sample based algorithms such as Rapidly-exploring Random Tree (RRT), Probabilistic Roadmap (PRM) and Rapidly-exploring Random Tree Star (RRT*) based approaches are the typical global path planning algorithms [2-4]. They are probabilistic complete and computationally efficient especially if the constraints are complex. In general, the global path planner and the local path planner are combined to generate a feasible path [5]. And the path achieves one or more conditions of optimal criteria such as shortest path, smoothness, safety and minimum energy consumption.

The continuous and differentiable local path planner is required for nonholonomic system such as car-like vehicle and motorcycle. The Dubins path investigated in [6], [7] generates a geometry curve consisted of circular arcs and line segments. Six different kinds of paths are computed and the shortest path is chosen as the Dubins path. The Reeds-Shepp (RS) path is introduced in [8] for the car that can move backward. However, the vehicles need to stop and reorient the steering wheels if the curvature of the path is discontinuous. In order to avoid the discontinuity, the continuous curvature (CC) path and the Elementary path is presented in [9], [10]. Wilde investigates the properties of the clothoid segments and an effective computation method for an Elementary path is presented [11], as well further studied in [12]. The Double Continuous-Curvature (DCC) path [13] is presented to converge to a line at the constant curvature rate. The local path planners based on the clothoid segments have been widely used in the car-like vehicles [5], [14] and aircrafts [15],[16] , this method has been deployed on the racing car [17]. Other typical curves like continuous cubic Bezier curves combined with RRT* [18] show the ability to generate a smooth and comfortable path. And a G^3-continuous path based on RS path is provided in [19]. However, large curvature and too many turns may occur in these methods and limit the speed of the vehicle [20] and cause unsafe maneuvering.

In practice, vehicles may suffer from uncertain, uneven, low friction and bumpy terrain, especially for motorcycles and bicycles on unstructured roads [21]. A large roll angle is required while the path curvature is large and may cause uncertain and unstable movements (such as skidding or capsizing) for the motorcycle, which is not expected. Thus the local path planner that generates a path of minimum curvature is required in this situation. Furthermore, the path generated from the sample based algorithms usually turns frequently, which can be seen in the result of [3] and [18]. It may have an unstable effect to the trajectory tracking algorithm. Thus the existing local path planners are not suitable for vehicles sensitive to roll angel and terrain conditions. In this paper, a novel local path planner based on Dubins path and CC path is proposed. The simulation shows that the performance of least maximum curvature and the shortest length is obtained by the path presented in this paper. The novel local path planner also shows the ability to combine with global path planner and provides a smoother path with less sharp turns.

This paper is organized as follows. The Double Continuous Curvature Turns (DCCT) path presented in this paper is introduced in section II and some related properties are provided as well. In section III, the computational method for the DCCT path is presented based on the analysis. The performances of the DCCT path are exposed and compared to other paths. Moreover, the novel local path planner combined with RRT* and is compared to CBB-RRT* shown in [18] in section IV. Finally, the conclusions and future perspectives are given in Section V that ends the paper.

II. Definitions and Properties

Firstly the definition of problem is presented in the following subsection. Then the path composed by two arcs is defined as Arc-Arc (AA) path. Finally, the Double Continuous Curvature Turns (DCCT) path based on the AA path and CC turns is presented. The properties discussed in this section are used for further analysis and investigation of the DCCT path in section III.

A. Problem Definition

A car-like vehicle model moving forward only with no skidding mentioned in [10] can be described as:

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$$\begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{\kappa} \end{pmatrix} = \begin{pmatrix} \cos\theta \\ \sin\theta \\ \kappa \\ 0 \end{pmatrix} v + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \sigma \tag{1}$$

where κ denotes the curvature and σ denotes the curvature change rate. The vector $q = (x, y, \theta, \kappa)$ is the state of the vehicle, where x and y are the coordinates of the model, and θ is the orientation of the model. The input vector $u = (v, \sigma)$ contains two variables: the driving speed of the rear wheels v, and the curvature rate σ. The curvature and the curvature rate are saturated to the mechanical structure of the vehicle:

$$|\kappa| \le \kappa_{\max}, \quad |\sigma| \le \sigma_{\max} \tag{2}$$

The path generated from the model (1) must be continuously differentiable and first order continuously differentiable. The main object of this paper is finding a feasible path from the original state $q_0 = (x_0, y_0, \theta_0, \kappa_0)$ to the target state $q_d = (x_d, y_d, \theta_d, \kappa_d)$ and satisfying the steering limitation. Suppose that $\kappa_0 = \kappa_d = 0$ and the problem is rotation invariance. The states can be rewritten as $q_0 = (0,0,\theta_0)$ and $q_d = (d, 0, \theta_d)$ where d denotes the Euclidean distance between two states.

B. AA(Arc-Arc) Paths

Dubins set D includes six admissible paths [7]. Considering two of them $\{C_L S C_R, C_R S C_L\} \in D$, where C_L, S, C_R denote turning left, going straight, and turning right. As the radius R of each arc increases until the two arcs are tangent, the path between q_0 and q_d consisting of two arcs can be determined, and each arc has the same radius:

- $\theta_0 + \theta_d < 0$, the path is a counterclockwise oriented arc of a circle of radius R following a clockwise oriented arc of a circle of radius R.
- $\theta_0 + \theta_d > 0$, the path is a clockwise oriented arc of a circle of radius R following a counterclockwise oriented arc of a circle of radius R.
- $\theta_0 = -\theta_d \ne 0$, the path is a counterclockwise oriented arc of a circle of radius $R(\theta_0 < 0)$ or a clockwise oriented arc of a circle of radius $R(\theta_0 > 0)$.
- $\theta_0 = -\theta_d = 0$, the path is a line segment.

Then the path designed above is called a AA(Arc-Arc) path and denoted as $AA(q_0, q_d)$. In general, $\theta_0 \ne -\theta_d$, the path is made up of two arcs with different orientation, and the orientation of the first arc is determined by $-sign(\theta_0 + \theta_d)$. Because the two arcs have a unique point of tangency, the AA path is unique.

Property 1. A AA path is a Dubins path if the orientations of the original state and the goal state satisfy one of the following conditions:

$$\begin{aligned} &\theta_0 \in (0, \pi/2), \theta_d \in (0, \pi/2) \\ &\theta_0 \in (-\pi/2, 0), \theta_d \in (-\pi/2, 0). \end{aligned} \tag{3}$$

Proof. Dubins path is the shortest path in the Dubins set D. As defined above, AA path is in the Dubins set. And according to the classification of the Dubins set in [7] , Dubins path has the form of $C_L S C_R$ or $C_R S C_L$ if the orientations of θ_0 and θ_d satisfy one of the conditions in (3). Then the AA path derived from the Dubins set is the Dubins path. □

Let $C = (x_C, y_C)$ be the coordination of the tangent point in the AA path, and the orientation of the point C is θ_C. Let $A = (x_0, y_0)$, $B = (x_d, y_d)$, O_1 be the circle center of the first arc and O_2 be the circle center of the second arc. Then define:

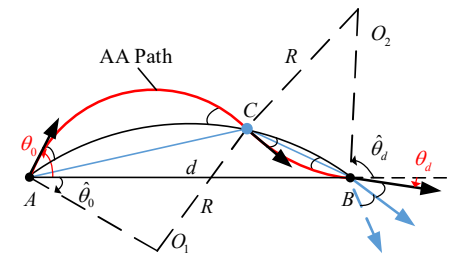

Fig.1. AA path

$$\hat{\theta}_0 = \theta_0 - sign(\theta_0 + \theta_d) \cdot \pi/2, \ \hat{\theta}_d = \theta_d + sign(\theta_0 + \theta_d) \cdot \pi/2 \tag{4}$$

where $\hat{\theta}_0$ is the angle between line AB and line AO_1, $\hat{\theta}_d$ denotes the angle between line AB and line BO_2 (see Fig.1).

Property 2. Geometric properties of the AA path.

1) The radius R of each arc of the AA path and the state of the point C $q_c = (x_c, y_c, \theta_c)$ can be derived from the geometrical relationship shown in Fig.1.

2) Tangent point C is on the same circular arc $\widehat{ACB}$ that goes through A and B if $\theta_0 - \theta_d = Const$. And the arc $\widehat{ACB}$ has orientation $(\theta_0 - \theta_d)/2$ at the point A.

Property 3. Define a collection of continuously differentiable paths $\mathcal{S}$ which starts at q_0, finishes at q_d and satisfies:

$$\max_{0 \le t \le T}(\theta(t)) - \min_{0 \le t \le T}(\theta(t)) \le \pi/2 \tag{5}$$

where $T > 0$ denotes the finish time of the path and suppose the start time of the path is 0. Then the AA path $AA(q_0, q_d)$ has the minimum curvature among all paths in $\mathcal{S}$ if it meets the condition (5).

Proof. Shown in Appendix A. □

Remark 1. The AA path is presented to derive the novel local path planner shown in the following subsection. Therefore the properties of the AA path can be used for analysis of the novel local path planner shown in section III.

C. DCCT(Double Continuous Curvature Turns) Paths

CC(Continuous Curvature) Path, CC Turn , and CC Circle are introduced in [10]. They are all made up of clothoid arcs, circular arcs and line segments. CC Path is more useful than Dubins path in practice because of the property of continuous curvature. Denote μ as the angle between the orientation of q_{out} and the tangent of the CC Circle at q_{out} (shown in Fig.2) where q_{out} is the end state of the CC Turn. If the deflection of the CC Turn δ_{CC} is not too small (i.e. $\delta_{CC} > \kappa^2/\sigma$, otherwise the CC Turn is an elementary path [10]), then μ can be calculated as:

$$\mu = \arctan\left(\frac{\sqrt{\frac{\pi}{\sigma}} C_f\left(\sqrt{\frac{\kappa^2}{\pi\sigma}}\right) - \sin\left(\frac{\kappa^2}{2\sigma}\right)/\kappa}{\sqrt{\frac{\pi}{\sigma}} S_f\left(\sqrt{\frac{\kappa^2}{\pi\sigma}}\right) + \cos\left(\frac{\kappa^2}{2\sigma}\right)/\kappa} \right) \tag{6}$$

where C_f and S_f are the Fresnel integrals and κ is the ending curvature of the clothoid in the CC Turn.

Notice that μ is constant if κ^2/σ is a constant. This property can also be derived from the theorem of clothoid segments in [11]. The theorem means that the shape of the clothoid segment is determined by the total deflection $\delta = \kappa^2/2\sigma$. So if κ^2/σ is a constant, the relationships in the clothoid can be obtained:

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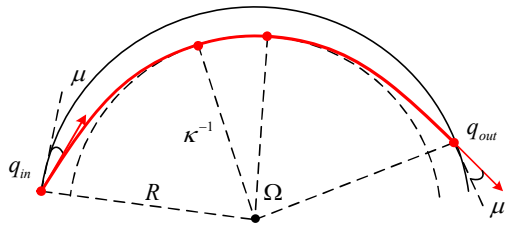

Fig. 2. CC Turn

$$\mu,\ \delta = Const \tag{7}$$

$$\kappa \propto \sqrt{\sigma},\ 1/\Delta x,\ 1/\Delta y,\ 1/R \tag{8}$$

where $\Delta x, \Delta y$ mean the forward distance and the lateral distance traveled in [11]. So the value of κ^2/σ can be called *shape factor*.

Property 4. Properties of the CC Turn.

1) The radius R and the angle μ of the CC Turn can be uniquely determined by giving the curvature $\kappa > 0$ and the curvature rate $\sigma > 0$: If κ and σ are determined, then μ can be calculated from the equation (6) uniquely and the center of the CC Circle Ω can also be calculated uniquely. Therefore the radius R is unique. Define the function f:

$$(\mu, R) = f(\kappa, \sigma). \tag{9}$$

2) If tuple (μ, R) is feasible (i.e. there are κ and σ that makes equation (9) hold), then the tuples $(\mu, R \in (0, \infty))$ are all feasible: If the equation (9) holds, one can conclude from (7) and (8) that $(\mu, R/k) = f(\kappa \cdot k, \sigma \cdot k^2)$ holds for any positive real number $k > 0$. Then all tuples $(\mu, R \in (0, \infty))$ can be hold.

3) There exists a maxmium angle μ_{max} that the tuples $(\mu \in (0, \mu_{max}), R \in (0, \infty))$ are all feasible: One can conclude from (6) that μ is a continuous function about κ and σ. It implies from the geometry of the CC Turn that $\mu \in (0, \pi/2)$, so the upper bound μ_{max} and the lower bound μ_{min} of μ must be existed. If one keep $\kappa = 1$ and vary the value of σ, then every value of μ can be obtained. As $\sigma \to \infty$, the lower bound $\mu_{min} \to 0$ can be obtained. So for all $\mu \in (0, \mu_{max})$, there are κ, σ, and R ensure the equation (9). Then one conclude from the property 4.2 that the tuples $(\mu \in (0, \mu_{max}), R \in (0, \infty))$ are all feasible.

Definition. A DCCT(Double Continuous Curvature Turns) path combines two CC Turns of different orientation to provide a solution of connecting the original state q_0 and the target state q_d. And each CC Circle has the same radius R and the same curvature rate σ. The orientations of the CC Circles are the same as the definition of the AA path.

Property 5. The DCCT path can be derived from the AA path. The AA path and the radius R of it can be determined if the angle $\mu \in (0, \mu_{max})$ is given. The AA path can be described as $AA(\tilde{q}_0, \tilde{q}_d)$, where:

$$\tilde{q}_0 = \left(0, 0, \tilde{\theta}_0\right),\ \ \tilde{q}_d = \left(d, 0, \tilde{\theta}_d\right), \tag{10}$$

$$\tilde{\theta}_0 = \theta_0 + sign(\theta_0 + \theta_d) \cdot \mu,\ \ \tilde{\theta}_d = \theta_d + sign(\theta_0 + \theta_d) \cdot \mu. \tag{11}$$

Here one can take the circle of the AA path as the CC Circle of the DCCT path (shown in Fig.3). Then there exists κ and σ that meet the equation (9).

Inference 1. The tangent point C of the DCCT path is the same as the tangent point in the AA path $AA(\tilde{q}_0, \tilde{q}_d)$.

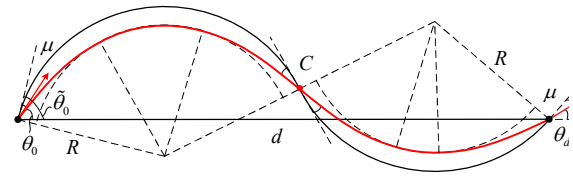

Fig. 3. DCCT path

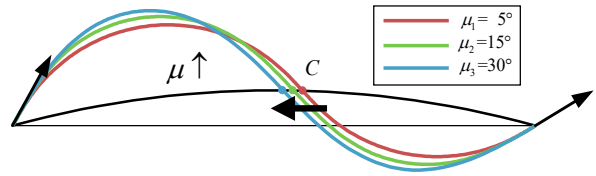

Fig. 4. The movement of the joining point C of the two CC Turns in the DCCT path with the variation of μ

Proof . The CC Circles of the DCCT path and the circles of the AA path $AA(\tilde{q}_0, \tilde{q}_d)$ are the same, so they have the same tangent point C. □

Inference 2. The tangent point C of the DCCT path is on the circular arc $\widehat{ACB}$ that defined in property 2 if $\theta_0 - \theta_d =$ Const.

Proof . Refer to the inference 1 and the property 2.2, inference 2 can be concluded directly. □

Remark 2. Property 4 shows the existence of the DCCT path. And the uniqueness of the DCCT path in general cases is presented in section III. Property 5 and inferences present the computational method which is further discussed in the following section.

III. Analysis and Calculation Method

The problem is to generate a DCCT path connecting the states q_0 and q_d. Four key variables need to be determined in order to solve the problem: μ, R, κ, σ. Firstly, the relationship between the four variables is analyzed. Then the properties of the DCCT path are further investigated. At last, an algorithm based on the properties and the analysis to compute the four variables of the DCCT path are provided.

A. Analyse

The variation of the angle μ denoted in (6) with respect to the curvature rate σ is plotted in Fig.4. And the curvature is fixed ($\kappa = 1$) as it just scale the horizontal axis. From Fig.4 the μ_{max} defined in property 4.3 can be obtained by numerical computation where $\mu_{max} \approx 48.37°$. In general, a crossover in a CC Turn is not expected, otherwise the planned path is unreasonable. Thus the deflection of the clothoid $\delta = \kappa^2/2\sigma$ must be smaller than $\pi - \mu$ yields:

$$\mu < \pi - \kappa^2/(2\sigma). \tag{12}$$

The numerical solution of the σ derived from (6) and (12) is unique and $\sigma \approx 0.218$ (shown in Fig.4). The first order derivative of μ is negative while $\mu = \pi - \kappa^2/2\sigma$, then the function (9) is bijective if the condition (12) is met. Furthermore, for the given original state and the target state, if the μ is determined, then the R, κ and σ are all unique under the conditions $\delta_{CC} > \kappa^2/\sigma$ and (12).

So the first step of the algorithm is to determine the angle μ, then the radius R of the AA path $AA(\tilde{q}_0, \tilde{q}_d)$ can be computed using the equation (4), (11). While μ and R are obtained, the κ^2/σ can be computed by using the inverse function of the equation (9) as it is bijective with the conditions $\delta_{CC} > \kappa^2/\sigma$ and (12):

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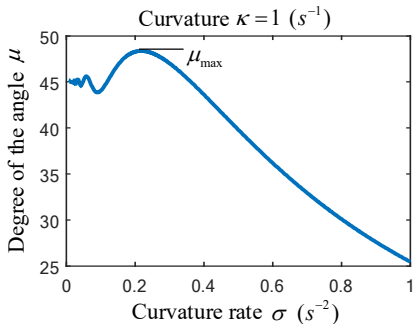

Fig. 5. The variation of the angle μ with respect to the curvature rate σ ($\kappa = 1$)

$$\kappa, \sigma = f^{-1}(\mu, R). \tag{13}$$

And define the function g and h from (13):

$$\kappa = g(\mu, R(\mu)), \quad \sigma = h(\mu, R(\mu)). \tag{14}$$

Inference 2 shows that the joining point of the two CC Turns in the DCCT path lies on a circular circle as the μ varies. So if there is a DCCT path connecting the states q_0 and q_d, then there is a unique joining point C and the point C moves toward one side as the μ increases (see Fig.5). And the derivative of the curvature rate σ with respect to μ can be derived from (14):

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\mu} = \frac{\partial h}{\partial \mu}\mathrm{d}\mu + \frac{\partial h}{\partial R}\frac{\partial R}{\partial \mu}\mathrm{d}\mu. \tag{15}$$

The first term on the right-hand side of the equation (15) mainly affects the shape of the CC Turns in the DCCT path. The second term mainly affects the radius of the CC Circles in the DCCT path. In practice, the value of (15) is negative while μ increases from 0 then positive while μ approach μ_{max}. Suppose there are κ and σ that satisfy the conditions (2) and (14), otherwise there is no feasible DCCT path. Hence, the angle μ can be searched quickly by dichotomy on the interval that satisfies:

$$\mathrm{d}\sigma / \mathrm{d}\mu < 0. \tag{16}$$

Consider solving the equation (13) at the given μ and R. Geometric conditions can be determined from Fig.6:

$$|R|\cos(\mu + \delta) = 1/\kappa - \Delta y. \tag{17}$$

$$R_\Omega = |R|. \tag{18}$$

where Ω denotes the center of the CC Circle generated by the variables κ and σ, R_Ω denotes the radius of the CC Circle. Firstly, the values of κ and σ are guessed as κ_0 and σ_0. Then the value of δ, Δy and R_Ω can be computed.

Notice that the equation (17) can always be hold if :

$$k = \frac{1/\kappa - \Delta y}{|R|\cos(\mu + \delta)} > 0 \tag{19}$$

and let the size of clothoid defined by κ_0 and σ_0 times the factor of $1/k$ (i.e. $\kappa_0 \times k$, $\sigma_0 \times k^2$, R_Ω/k, $\Delta y/k$). Then the equation (17) is met while the shape of the clothoid holds. Hereafter (18) is used to check whether the shape of the clothoid satisfies the condition. So the computation of (13) can be derived quickly using dichotomy.

B. Properties of the DCCT path

Suppose that the DCCT path exists between the original state and the target state. Denote κ^{max} as the maximum curvature in the path. An important property of the DCCT path can be concluded that the minimum κ^{max} can be obtained if $|\sigma| = \sigma_{max}$ and the condition holds:

$$\tilde{\theta}_0 - \angle CAB < \pi/2. \tag{20}$$

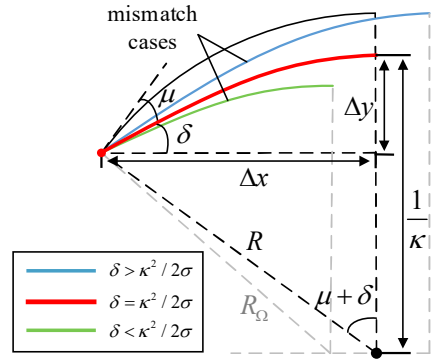

Fig. 6. Geometry conditions for κ, σ at the given μ and R

In addition, the AA path $AA(\tilde{q}_0, \tilde{q}_d)$ which derives the DCCT path meets the condition (5).

Property 5 shows that the DCCT path can be derived from the AA path $AA(\tilde{q}_0, \tilde{q}_d)$. Then property 3 shows that if the AA path meets the condition (5), then the radius of the CC Circles of the DCCT path are maximum. In general, the condition (20) holds, otherwise the states are required to be replanned. Then the values of radius R and σ decreases with respect to the increasement of μ. So as μ increases from 0, σ decreases to meet the maximum curvature rate constraint. Then the minimum μ and the max radius R are obtained. The angle μ is proportional to the *shape factor* κ^2/σ with the condition (12) as Fig.4 shows. Suppose $\mu_1 < \mu_2 < \mu_{max}$, and one has $(\kappa_1, \sigma_1) = f^{-1}(\mu_1, R)$, $(\kappa_2, \sigma_2) = f^{-1}(\mu_2, R)$, then the result $\kappa_1 < \kappa_2$ can be proved easily. It can be proved that for a smaller μ the DCCT path has smaller κ^{max} and larger R with the conditions mentioned above. Furthermore, if the DCCT path does not contain elementary paths, then the DCCT path with the conditions mentioned above has the minimum κ^{max} among all feasible CC Paths connecting the original state and the target state and satisfying the condition (5).

C. Computing the DCCT path

The DCCT path construction process is shown in Algorithm 1. The input of the algorithm is the original state q_0, the target state q_d, and the constrains shown in (2). The number of iterations i, the lower and the upper bounds of μ are initialized as 0, 0 and $\pi/4$ (Slightly less than the maximum μ analyzed in III.A). And the accuracy of μ derived from the algorithm is denoted as ϵ. The algorithm firstly normalizes the input (Line 3). Then the AA path determined by q_0, q_d and μ can be computed (Line 5). The function GetCurvature() shown in Algorithm 2 solves the equation (13). The equation (12) is satisfied in lines 7-8. Then the function Update() updates the variable μ until the set accuracy ϵ is reached, and the boundary of μ is updated using dichotomy. Finally, the DCCT path can be generated using the function GenerateDCCTPath() as all parameters to compose the two CC Circles are determined (the principle of the steering method is shown in [10]).

In Algorithm 2, the parameters sf_{low} and sf_{up} are denoted as the lower bound and the upper bound of the *shape factor* κ^2/σ. The deflection and the distance traveled of the clothoid can be computed (Line 3 [11]). The radius R_Ω can be computed as:

$$R_\Omega = \|(\Delta x, \Delta y - 1/\kappa)\|_2.$$

The factor k shown in (19) is used to scale the clothoid (Lines 4-6). After that, the shape of the clothoid is adjusted by

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dichotomy (Lines 8-21). The coefficient k_σ is greater than 1 and is usually 2.

Algorithm 1: DCCT path Generation

1 **Require**: $q_0, q_d, \kappa_{max}, \sigma_{max}$
2 **Initialize**: $i = 0,\ \mu_{min} = 0,\ \mu_{max} = \pi/4,\ \mu_i,\ \epsilon$
3 $\theta_0, \theta_d, d \leftarrow$ NormalizeInput(q_0, q_d)
4 **repeat**:
5 $R_i, q_c \leftarrow$ GenerateAAPath $(\theta_0 + \mu_i, \theta_d + \mu_i, d)$
6 $\kappa_i, \sigma_i \leftarrow$ GetCurvature $(\mu_i, R_i, \sigma_{max})$
7 **If** $\kappa_i^2/2\sigma_i + \mu_i > \pi$ **then**
8 **return** failed
9 $\mu_{i+1},\ \mu_{min},\ \mu_{max} \leftarrow$ Update$(\mu_i,\ \mu_{min},\ \mu_{max}, \sigma_i, \sigma_{max})$
10 **If** $\mu_{max} - \mu_{min} < \epsilon$ **then**
11 **If** $|\kappa_i| > \kappa_{max}$ **then**
12 **return** failed
13 **If** $|\sigma_i| > \sigma_{max}$ **then**
14 **return** GenerateDCCTPath$(q_0, q_d, q_c, \mu_{min}, R_i, \kappa_i, \sigma_i)$
15 **return** GenerateDCCTPath$(q_0, q_d, q_c, \mu_i, R_i, \kappa_i, \sigma_i)$
16 i++

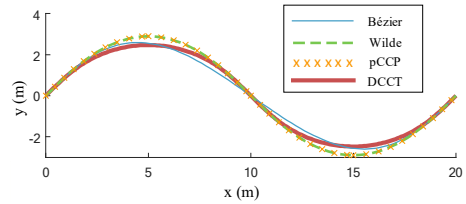
Algorithm 2: GetCurvature$(\boldsymbol{\mu}, \boldsymbol{R}, \boldsymbol{\sigma_{max}})$

1 **Initialize**: $i \leftarrow 0, \kappa_i \leftarrow 1/R, \sigma_i \leftarrow \sigma_{max}, sf_{low} \leftarrow 0, sf_{up} \leftarrow inf, \epsilon$
2 **repeat**:
3 $\Delta x, \Delta y, \delta \leftarrow$ GetClothoid(κ_i, σ_i)
4 $k \leftarrow$ GetScaleFactor$(\mu, \kappa_i, \Delta y, \delta)$
5 **If** $k > 0$ **then**
6 $\Delta x, \Delta y, \kappa_i, \sigma_i \leftarrow \Delta x/k, \Delta y/k, \kappa_i \cdot k, \sigma_i \cdot k^2$
7 $R_\Omega \leftarrow$ GetCircleRadius$(\Delta x, \Delta y, \kappa_i)$
8 **If** $k > 0$ **and** $(\mu + \delta - \pi/2) \cdot (R_\Omega - R) \leq 0$ **or**
9 $k < 0$ **and** $\mu + \delta \leq \pi/2$
10 $sf_{up} \leftarrow \kappa_i^2 \cdot \sigma_i$
11 **If** $sf_{low} = 0$ **then**
12 $\sigma_i \leftarrow k_\sigma \cdot \sigma_i$
13 **Else**
14 $\sigma_i \leftarrow (\kappa_i^2/\, sf_{low} + \kappa_i^2/sf_{up})/2$
15 **If** $k > 0$ **and** $(\mu + \delta - \pi/2) \cdot (R_\Omega - R) > 0$ **or**
16 $k < 0$ **and** $\mu + \delta > \pi/2$
17 $sf_{low} \leftarrow \kappa_i^2 \cdot \sigma_i$
18 **If** $sf_{up} = inf$ **then**
19 $\sigma_i \leftarrow \sigma_i/k_\sigma$
20 **Else**
21 $\sigma_i \leftarrow (\kappa_i^2/\, sf_{low} + \kappa_i^2/sf_{up})/2$
22 **If** $sf_{up} - sf_{low} < \epsilon$ **then**
23 **return** κ_i, σ_i
24 i++

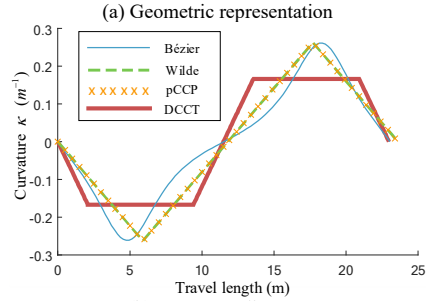
IV. Example Application

The lane change maneuver is a typical maneuver mode of the vehicles. Suppose that the original state is $q_0 = (0,0,\pi/4,0)$, the target state is $q_d = (20,0,\pi/4,0)$ and the maximum curvature rate is $|\kappa| < 0.08$. The performances of the DCCT path and three existing paths are shown in Fig.7. The fifth order Bézier curve is used to ensure the original and target orientations and curvatures. The Wilde's path [11] is composed by four clothoid. And the parametric Continuous Curvature Path (pCCP [14]) records the same result as the Wilde's path in the lane change maneuver.

The comparison results are summarized in Table I for the lane change curves. The curvature rate is least for the Wilde's path and pCCP. The Bézier curve has large curvature and curvature rate, therefore it may be hard to use in practice. The DCCT path records the least curvature while the maximum curvature rate is achieved. Less curvature also means that less steering angle is required and higher speed can be archived. The results given in the Table I also shows that the DCCT path provides a shortest path. Thus the DCCT path takes less time to travel with limited curvature rate. In addition, the algorithms represented in the section III have the ability of real-time computing. Therefore the DCCT path can be combined with global path planner such as RRT* based algorithms.



(a) Geometric representation



(b) Curvature diagram

Fig. 7. Lane change paths and curvatures of the four methods.

TABLE I. Paths Comparison of the Four Methods

Method	$max\ \kappa$	$min\ \kappa$	$max\ \sigma$	$min\ \sigma$	*length*
Bézier	0.2615	-0.2615	0.0671	-0.0724	23.04
Wilde	0.2627	-0.2627	0.0445	-0.0445	23.60
pCCP	0.2627	-0.2627	0.0445	-0.0445	23.60
DCCT	**0.1666**	**-0.1666**	**0.0800**	**-0.0800**	**22.98**

To show the impact of the DCCT path combined with RRT*, the DCCT path generated by smoothing the path generated by RRT* is compared with CBB-RRT* [18]. The result of simulation shows that fewer sharp turns occur in the DCCT path as we can see in Fig.8. Therefore, the path with smaller maximum curvature is obtained. Furthermore, a safer and smoother path with faster speeds can be achieved by our novel local path planner.

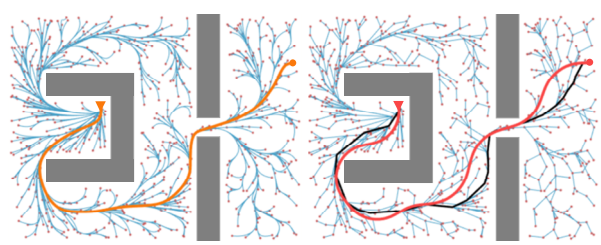


Fig. 8. CBB-RRT* and DCCT path combined with RRT*. Orange path in the left-hand image represents the path generated from the CBB-RRT*. Red path in the right-hand image represents the DCCT path, and black path represents the path generated from RRT*.

V. Conclusion

This paper presents a new clothoid based local path planner named the DCCT path which is composed with two CC Turns. The least maximum curvature can be obtained with limited curvature rate if the elementary path is not contained in the DCCT path. In addition, the paper discusses the computational method with dichotomy for the DCCT path. The examples of lane change maneuver and combination with RRT* are presented. The smooth, feasible and drivable paths

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with less sharp turns can be generated in real-time (the algorithm runs on a laptop with 16 GB of RAM and an Intel i7-8700 processor running MATLAB R2019b and costs 0.335s on average for a single DCCT). And our method has a minimal amount of steering (maximum curvature) as well as shortest length of path compared with existing methods shown in the simulation. As a future work, the combination with global path planners will be further investigated to solve the problem of path planning on complex terrain.

APPENDIX A

The AA path has four kinds of orientation curves (see Fig.A1). Here we proof the first kind of them, the others have the similar process of proofs.

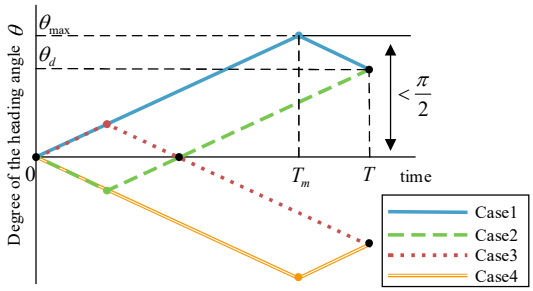

Fig. A1. Orientation diagram of the AA paths

Suppose that the AA path connects the states q_0 with q_d and the original orientation θ_0 is zero. The AA path gets the maximum orientation θ_{max} at time T_m and end at time T with the orientation θ_d. Suppose a curve $\tilde{S}$ connecting the states q_0 and q_d exists with smaller curvature. Then it leads to:

$$\int_0^T \cos\theta(\tau)d\tau = \int_0^{\tilde{T}} \cos\tilde{\theta}(\tau)d\tau, \tag{A.1}$$

$$\int_0^T \sin\theta(\tau)d\tau = \int_0^{\tilde{T}} \sin\tilde{\theta}(\tau)d\tau, \tag{A.2}$$

$$\theta(T) = \tilde{\theta}(\tilde{T}) = \theta_d, \tag{A.3}$$

$$\max(d\theta(t)/dt) > \max(d\tilde{\theta}(\tilde{t})/d\tilde{t}).$$

where $\tilde{T}$ denotes the end time of the $\tilde{S}$. $\theta(t)$ and $\tilde{\theta}(t)$ are the orientations of the AA path and the path $\tilde{S}$ with respect to t. We can conclude easily from (A.3) and (5) that:

$$\pi/2 > \theta(t) > \left|\tilde{\theta}(t)\right|, t \in (0, T_m). \tag{A.4}$$

Thus, the following inequalities are derived:

$$\int_0^{T_m} \cos\theta(\tau)d\tau < \int_0^{T_m} \cos\tilde{\theta}(\tau)d\tau, \int_0^{T_m} \sin\theta(\tau)d\tau > \int_0^{T_m} \sin\tilde{\theta}(\tau)d\tau. \tag{A.5}$$

The discussion can be divided into three cases:

1) If the crossover between the two orientation curves exists. It can only happen at time $t_c \in (T_m, T)$. Then it implies that $\theta(t_c) = \tilde{\theta}(t_c)$ and the condition $T < \tilde{T}$ can be derived from (A.3). Thus, the following inequality holds:

$$\int_{t_c}^{T} \cos\theta(\tau)d\tau < \int_{t_c}^{T} \cos\tilde{\theta}(\tau)d\tau + \int_{T}^{\tilde{T}} \cos\tilde{\theta}(\tau)d\tau. \tag{A.6}$$

The condition (A.1) is violated.

2) If there is no crossover between the two orientation curves and $T \geq \tilde{T}$, then we have:

$$\int_0^{\tilde{T}} \sin\theta(\tau)d\tau + \int_{\tilde{T}}^{T} \sin\theta(\tau)d\tau > \int_0^{\tilde{T}} \sin\theta(\tau)d\tau > \int_0^{\tilde{T}} \sin\tilde{\theta}(\tau)d\tau \tag{A.7}$$

The condition (A.2) is violated.

3) If there is no crossover between the two orientation curves and $T < \tilde{T}$, then one has:

$$\int_0^{T} \cos\theta(\tau)d\tau < \int_0^{T} \cos\tilde{\theta}(\tau)d\tau < \int_0^{T} \cos\tilde{\theta}(\tau)d\tau + \int_{T}^{\tilde{T}} \cos\tilde{\theta}(\tau)d\tau. \tag{A.8}$$

The condition (A.1) is violated.

Above all, the curve $\tilde{S}$ does not exist. It is proved by contradiction that the AA path have the minimum curvature under the condition (5).

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